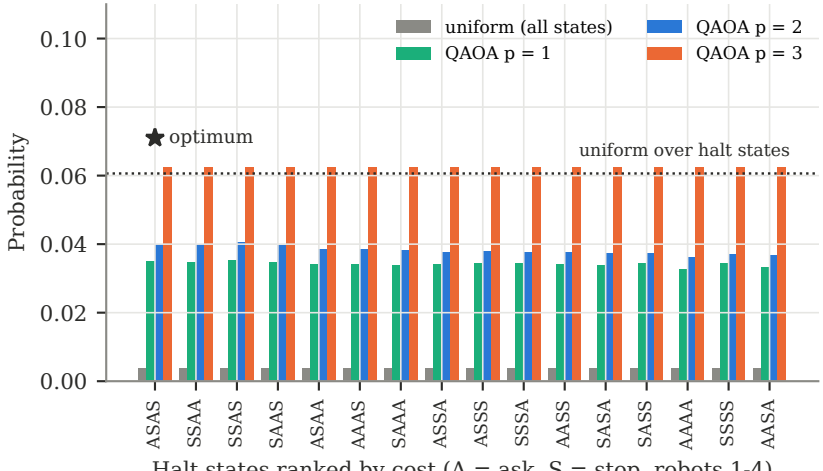
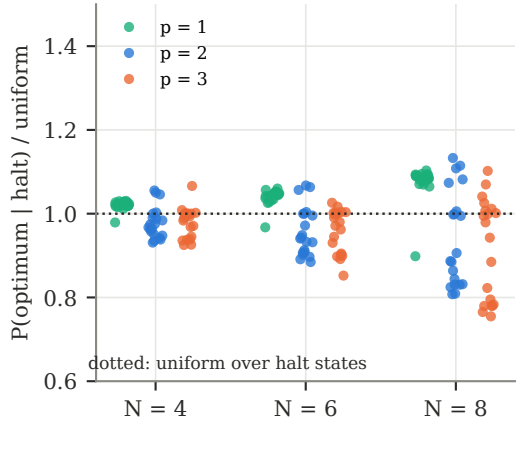

(a) One N = 4 instance: the 16 states where every robot halts
uniform (all states)
QAOA p = 1
QAOA p = 2
QAOA p = 3
optimum
uniform over halt states
Probability
0.10
0.08
0.06
0.04
0.02
0.00
ASAS
SSAA
SSAS
SAAS
ASAA
AAAS
SAAA
ASSA
ASSS
SSSA
AASS
SASA
SASS
AAAA
SSSS
AASA
Halt states ranked by cost (A = ask, S = stop, robots 1-4)


(b) All 60 final instances
p = 1
p = 2
p = 3
P(optimum | halt) / uniform
1.4
1.2
1.0
0.8
0.6
dotted: uniform over halt states
N = 4
N = 6
N = 8